

1. Administrative & Study Identification

Full Study Title: A Phase IIb, Open-label, Single-arm, Global Study of Perioperative Durvalumab With Neoadjuvant ddMVAC or Gem/Cis in Patients With Muscle-invasive Bladder Cancer (NIAGARA-2)
Short Title/Acronym: NIAGARA-2 **Protocol Number:** D933RC00002 **NCT Number:** NCT06960577 **Posting ID:** 358562189799662294 **Sponsor/Company Name:** AstraZeneca **Trial Status:** Recruiting
Investigational Product: Durvalumab in combination with investigator-selected chemotherapy (ddMVAC [Methotrexate, Vinblastine, Doxorubicin, Cisplatin] or Gemcitabine/Cisplatin)
Disease Areas/Interventions: Urinary Bladder Neoplasms, Immune Checkpoint Inhibitors, Methotrexate, Vinblastine, Doxorubicin, Cisplatin (ATC: L01XA01, Trade Name: Platinol), Gemcitabine **Phase of Development:** Phase III / Phase IIb **Medical Monitor:** AstraZeneca Clinical Study Information Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety of neoadjuvant durvalumab combined with ddMVAC or gemcitabine/cisplatin prior to radical cystectomy (RC) in patients with muscle-invasive bladder cancer (MIBC), specifically looking at the incidence of Grade 3 or 4 possibly treatment-related adverse events. **Secondary Objectives:** To evaluate the safety and tolerability of perioperative durvalumab combined with ddMVAC or gem/cis, and to evaluate the efficacy of perioperative durvalumab combined with ddMVAC or gem/cis in terms of event-free survival (EFS), disease-free survival (DFS), and overall survival (OS).

2.2 Background & Rationale Historically, cisplatin-eligible patients with MIBC have been offered neoadjuvant cisplatin-based chemotherapy followed by surgery, yet relapse rates remain high. The Phase 3 NIAGARA trial demonstrated significant improvements in EFS and OS by adding durvalumab to neoadjuvant chemotherapy followed by adjuvant durvalumab. The Phase IIb NIAGARA-2 study aims to expand on these data by investigating the same perioperative durvalumab regimen in combination with investigator-selected cisplatin-based chemotherapy in a broader clinical practice setting.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label (None) **Randomization:** N/A (Single Group Assignment) **Trial Status:** Recruiting

3.2 Planned Enrollment & Duration Target **\$N\$:** 150 patients **Number of Sites:** Global, multi-center **Estimated Study Start:** 15-May-2025 **Estimated Primary Completion:** 31-Oct-2028

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria** 1. Age ≥ 18 years. 2. Participants with clinical tumour stage T2-T4aN0/1M0 or T1N1M0 with transitional or mixed transitional cell histology. 3. Patients must be planning to undergo radical cystectomy. 4. Patients who have not received prior systemic chemotherapy or immunotherapy for the treatment of muscle-invasive bladder cancer. 5. ECOG performance status of 0 or 1. 6. Minimum life expectancy of 12 weeks at the first dose of study medication.
- 4.2 Exclusion Criteria** 1. Evidence of lymph node (N2-N3) or metastatic (M1) disease. 2. Inoperable tumour(s) with fixation to the pelvic wall on clinical examination. 3. Prior exposure to immune-mediated therapy including, but not limited to, other anti-CTLA-4, anti-PD-1, anti-PD-L1, and anti-PD-L2 antibodies (excluding prior Bacillus Calmette-Guérin). 4. Current or prior use of immunosuppressive medication within 14 days before the first dose of durvalumab. 5. Any concomitant medication known to be contraindicated to the chemotherapy (ddMVAC or gem/cis). 6. Uncontrolled intercurrent illness.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:** Perioperative durvalumab combined with investigator's choice of standard neoadjuvant chemotherapy (ddMVAC or Gemcitabine/Cisplatin), followed by radical cystectomy, and subsequent adjuvant durvalumab monotherapy. **Route of Administration:** Intravenous (IV). **Duration of Treatment:** Neoadjuvant phase prior to surgery + adjuvant phase post-surgery (span of study monitoring up to 181 weeks).
- 5.2 Concomitant & Prohibited Therapy Permitted:** Prior intravesical Bacillus Calmette-Guérin (BCG) for non-muscle invasive disease is permitted. Standard antiemetics and hydration for cisplatin-based chemotherapy. **Prohibited:** Immunosuppressive medications within 14 days of start; concurrent systemic immunotherapies or other investigational anticancer therapies.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to 0 Informed Consent, ECOG Assessment, Baseline Imaging/Staging, Lab Tests Baseline
Day 0		Confirmation of Eligibility, First Dose of Neoadjuvant	

Treatment | | **Interim** | Pre-RC & Post-RC | Pre-RC: Safety Assessments (Grade 3/4 AEs), Surgery (Radical Cystectomy). Post-RC: Adjuvant Durvalumab, DFS/EFS/OS Monitoring | | **End of Study** | Week 181 | Final Follow-up, Survival Status, Final AE/SAE Assessment |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The safety of neoadjuvant durvalumab combined with ddMVAC or gem/cis prior to radical cystectomy (RC), specifically the incidence of Grade 3 or 4 possibly treatment-related adverse events (PRAEs) as observed prior to RC. Time frame: Up to 6 months. **Measurement Method:** Clinical safety assessments graded per NCI CTCAE criteria.

7.2 Secondary & Exploratory Endpoints

- * **Safety and Tolerability:** Evaluate the overall safety and tolerability of perioperative durvalumab combined with ddMVAC or gem/cis.
- * **Event-Free Survival (EFS):** The efficacy of perioperative durvalumab combined with ddMVAC or gem/cis measured by the time from first neoadjuvant treatment until first recurrence after RC, progression precluding RC, failure to undergo RC, or death.
- * **Disease-Free Survival (DFS):** The efficacy of perioperative durvalumab combined with ddMVAC or gem/cis measured by the time from RC to the earliest first recurrence of disease post-RC or death.
- * **Overall Survival (OS):** Time from first neoadjuvant treatment until death due to any cause.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring for the incidence, severity, nature, seriousness, and causality of all treatment-emergent adverse events (TEAEs), with special attention to immune-mediated adverse events (imAEs). **Serious Adverse Events (SAEs):** Routine reporting per sponsor and regulatory guidelines (typically 24 hours). Evaluated up to 90 days following the last dose of durvalumab. **Data Monitoring Committee (DMC):** Internal/External safety review committees governing trial safety data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all patients who receive at least one dose of study medication); Intent-to-Treat (ITT) Population for efficacy (DFS, EFS, OS). **Sample Size Justification:** $N=150$ is determined to be sufficient to estimate the safety profile (Grade 3/4 PRAEs) and to confirm the generalizability of the EFS/OS benefits previously observed in the phase 3 NIAGARA trial in a real-world clinical setting. **Handling of Missing Data:** Time-to-event outcomes (EFS, DFS, OS) will be summarized using Kaplan-Meier methodology. Patients

without an event at the time of analysis will be censored at the date of their last evaluable disease assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Periodic on-site and remote monitoring visits by AstraZeneca to ensure protocol compliance and data integrity.

Audit Trail: eCRF entry, source data verification, and standard query resolution via data clarification forms.

11. Study Identifiers & Governance

Primary ID: D933RC00002 **Registry Number:** NCT06960577 **Posting ID:** 358562189799662294 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** Perioperative Durvalumab With Neoadjuvant ddMVAC or Gemcitabine/Cisplatin in Patients With Muscle-invasive Bladder Cancer (NIAGARA-2) **Official Regulatory Title:** A Phase IIIB, Open-label, Single-arm, Global Study of Perioperative Durvalumab With Neoadjuvant ddMVAC or Gem/Cis in Patients With Muscle-invasive Bladder Cancer (NIAGARA-2)

12. Clinical Framework (Bladder Cancer Specific)

Primary Condition: Muscle-invasive Bladder Cancer (MIBC) / Urinary Bladder Neoplasms **MeSH Code:** D001749 **Preferred UMLS Name:** Chemoimmunotherapy **Diagnosis Threshold:** Clinical tumour stage T2-T4aN0/1M0 or T1N1M0 with transitional or mixed transitional cell histology. **Medical Methodology:** Perioperative Checkpoint Inhibition (Durvalumab) + Neoadjuvant Chemotherapy + Radical Cystectomy. **Optimization Target:** Neoadjuvant safety optimization and extending Event-Free / Disease-Free Survival.

13. Study Procedures & Methodology

Management Pathway: Neoadjuvant Chemo-Immunotherapy → Radical Cystectomy → Adjuvant Immunotherapy.

Education Protocol (HCP): Site initiation visits and PI training focusing on immune-mediated adverse event management and standard surgical clearance parameters. **Education Protocol (Patient):**

Informed consent detailing surgical and immunological risks.

Quality Audit Mechanism: Independent central and local tracking of AE/SAE reporting and treatment administration adherence.

14. Observation & Follow-Up Schedule

Total Duration: Up to ~181 weeks **Onsite Visit Cadence:** Matched to treatment administration cycles (typically Q2W/Q3W or Q4W during adjuvant therapy). **Remote Monitoring Frequency:** As required per site-specific monitoring plans. **Checklist Review Interval:** Pre-infusion assessments for immune-mediated AEs and laboratory clearances prior to each cycle.

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				